

MATH 6241 – Stochastic Processes

Section 3. Transitions

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- Transition probability
- Multi-step transitions
- Chapman-Kolmogorov Equations

□ Transition probability

Let $\{X_t\}$ be a Markov Chain with state space Ω .

Recall the (one-step) transition probability for homogenous Markov chains

$$p_{ij} := P(X_{n+1} = j \mid X_n = i) = \dots = P(X_1 = j \mid X_0 = i)$$

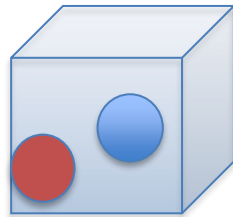
Denote the *initial distribution* row vector as $\alpha = (\alpha_1, \dots, \alpha_m)$ with

$$\alpha_i := P(X_0 = i)$$

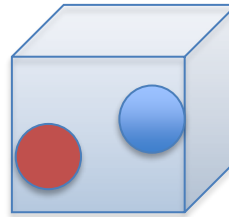
The joint probability is

$$P(X_0 = i, X_1 = j) = P(X_1 = j \mid X_0 = i)P(X_0 = i) = p_{ij} \alpha_i$$

Continue Example (Balls and Boxes Markov Chain) .



Box1



Box2

One ball is selected uniformly at random from each box; exchanged and then placed back into the boxes.

Let X_n denote the number of red balls in the first box after n steps.

Systematically:

p_{ij} = transition probability $i \rightarrow j$, where $i, j \in \Omega = \{0, 1, 2\}$ are possible states.

Transition Matrix

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{bmatrix}$$

(c) $\mathbb{P}(X_3 = 2 \mid X_0 = 1)$

Calculate: $\mathbb{P}(X_2 = j | X_0 = i)$



$$\mathbb{P}(X_2 = j | X_0 = i) = P(i \rightarrow^* \rightarrow j)$$

$$\mathbb{P}(X_2 = j | X_0 = i) = (P^2)_{ij}$$

The (i, j) entry of P^2 is

$$(P^2)_{ij} = \sum_k p_{ik} p_{kj} = \mathbb{P}(X_2 = j | X_0 = i)$$

Probabilistic meaning: P^2 gives two-step transition probabilities.

$$X_0 \rightarrow X_2 \rightarrow X_4 \rightarrow X_6 \rightarrow \dots$$

or
$$X_1 \rightarrow X_3 \rightarrow X_5 \rightarrow X_7 \rightarrow \dots$$

Each subsequence is itself a Markov chain, and its transition matrix is P^2

$$P^2 = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{8} & \frac{3}{4} & \frac{1}{8} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \end{pmatrix}.$$

For example,

$$(P^2)_{12} = \mathbb{P}(X_2 = 2 | X_0 = 1) = \frac{1}{4}$$

In a similar way, we “guess” the matrix P^k is the transition for the k -step chains

$$X_0 \rightarrow X_k \rightarrow X_{2k} \rightarrow X_{3k} \rightarrow X_{4k} \rightarrow \dots$$

That is
$$(P^k)_{ij} = \mathbb{P}(X_k = j | X_0 = i)$$

As in previous example,

$$\mathbb{P}(X_4 = 2 | X_0 = 0) = (P^4)_{02} =$$

$$P^4 = \begin{pmatrix} \frac{3}{16} & \frac{5}{8} & \frac{3}{16} \\ \frac{5}{32} & \frac{11}{16} & \frac{5}{32} \\ \frac{3}{16} & \frac{5}{8} & \frac{3}{16} \end{pmatrix}$$

The k -step Transition Probability

$$X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_{k-1} \rightarrow X_k \rightarrow X_{k+1} \rightarrow \cdots \rightarrow X_{2k} \rightarrow \cdots \rightarrow X_{3k} \rightarrow$$

The **k -step transition** probability $p_{ij}(k)$ (or $p_{ij}^{(k)}$) is the probability that a system currently in state " i " will transition to state " j " after " k " steps have passed,

$$p_{ij}(k) = \mathbb{P}(X_k = j | X_0 = i) = \mathbb{P}(X_{n+k} = j | X_n = i) \text{ for any } n$$

Theorem: [Chapman-Kolmogorov Equation] The k -step transition probabilities are given by the transition matrix P^k for any k :

$$p_{ij}(k) = (P^k)_{ij}$$

Remark: $p_{ij} = 0$ means that the chain cannot move from state i to state j in one step.

However, it is possible in this situation that there is an integer k such that $p_{ij}(k) > 0$, meaning that it is possible to move from i to j in k steps. In this case we say that state j is **accessible** from state i .

Proof: For $k = 2$

$$\begin{aligned} P(X_2 = j | X_0 = i) &= \sum_l P(X_2 = j, X_1 = l | X_0 = i) \\ &= \sum_l P(X_2 = j | X_1 = l, X_0 = i) P(X_1 = l | X_0 = i) \\ &= \sum_l P(X_2 = j | X_1 = l) P(X_1 = l | X_0 = i) \\ &= \sum_l p_{lj} p_{il} = \sum_l p_{il} p_{lj} = (P^2)_{ij} \end{aligned}$$

- Matrix power property

$$P^{n+m} = P^n P^m$$

The (i, j) -th entry

$$(P^{n+m})_{ij} = \sum_k (P^n)_{ik} (P^m)_{kj}$$

- Equivalently, in transition probability, we have

$$p_{ij}(n+m) = \sum_k p_{ik}(n) p_{kj}(m)$$

$$P(X_{n+m} = j | X_0 = i) = \sum_k P(X_n = k | X_0 = i) P(X_{n+m} = j | X_n = k)$$

The last equation can be proved directly:

$$p_{ij}(n+m) = P(X_{n+m} = j | X_0 = i)$$

$$= \sum_k P(X_{n+m} = j, X_n = k | X_0 = i)$$

$$= \sum_k P(X_{n+m} = j | X_n = k, X_0 = i) P(X_n = k | X_0 = i)$$

$$= \sum_k P(X_n = k | X_0 = i) P(X_{n+m} = j | X_n = k)$$

$$= \sum_k P(X_n = k | X_0 = i) P(X_m = j | X_0 = k)$$

$$= \sum_k p_{ik}(n) p_{kj}(m)$$

Example.

Consider the two-state weather Markov chain (Rainy/Sunny). If $\alpha = 0.7$ and $\beta = 0.4$, find the probability that it will **rain four days from today**, given that it is **raining today**.

The transition matrix (states ordered as R, S) is

$$P = \begin{pmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{pmatrix}.$$

Compute powers:

$$P^2 = \begin{pmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{pmatrix}, \quad P^4 = (P^2)^2 = \begin{pmatrix} 0.5749 & 0.4251 \\ 0.5668 & 0.4332 \end{pmatrix}.$$

Starting from Rainy today means the desired probability is the $(R \rightarrow R)$ entry:

$$\mathbb{P}(X_4 = R \mid X_0 = R) = (P^4)_{RR} = 0.5749.$$

Example (Two-day memory model as a 4-state Markov chain)

In the 2-day-memory weather model (states are the last two days),

The transition matrix under state space (RR, NR, RN, NN)

$$P = \begin{pmatrix} 0.7 & 0 & 0.3 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0.2 & 0 & 0.8 \end{pmatrix}$$

Given that it **rained on Monday and Tuesday**, what is the probability that it will **rain on Thursday**?

$$P^2 = \begin{pmatrix} 0.49 & 0.12 & 0.21 & 0.18 \\ 0.35 & 0.20 & 0.15 & 0.30 \\ 0.20 & 0.12 & 0.20 & 0.48 \\ 0.10 & 0.16 & 0.10 & 0.64 \end{pmatrix}.$$

$$P(\text{Thu} = R \mid \text{Mon} = R, \text{Tue} = R) = (P^2)_{RR,RR} + (P^2)_{RR,NR} = 0.49 + 0.12 = 0.61.$$

Example: Gambler's ruin

Assume that the gambler's initial stake is \$3 and the gambler plays until either **gaining \$6** or going bust.

At each play the gambler wins \$1 with probability 0.6, or loses \$1 with probability 0.4. Find the gambler's expected fortune after four plays.

State Space: $S = \{0,1,2,3,4,5,6\}$.

The transition matrix P is:

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.4 & 0 & 0.6 & 0 & 0 & 0 & 0 \\ 0 & 0.4 & 0 & 0.6 & 0 & 0 & 0 \\ 0 & 0 & 0.4 & 0 & 0.6 & 0 & 0 \\ 0 & 0 & 0 & 0.4 & 0 & 0.6 & 0 \\ 0 & 0 & 0 & 0 & 0.4 & 0 & 0.6 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

The fourth power P^4

$$P^4 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.4096 & 0 & 0.3456 & 0 & 0.1296 & 0 & 0 \\ 0.1536 & 0 & 0.4352 & 0 & 0.3456 & 0 & 0.0656 \\ 0 & 0.1536 & 0 & 0.4352 & 0 & 0.3456 & 0.0656 \\ 0 & 0 & 0.1536 & 0 & 0.4352 & 0 & 0.4112 \\ 0 & 0 & 0 & 0.1536 & 0 & 0.4352 & 0.4112 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Starting from state 3,

$$\pi_0 = (0,0,0,1,0,0,0)$$

$$\pi_4 = \pi_0 P^4 = (0,0.1536,0,0.4352,0,0.3456,0.0656).$$

Expected fortune:

$$E[X_4] = 1(0.1536) + 3(0.4352) + 5(0.3456) + 6(0.0656) = 3.744.$$

Joint Distribution

The joint probability:

$$P(X_0 = i, X_1 = j) = P(X_1 = j | X_0 = i)P(X_0 = i) = P(X_0 = i)p_{ij}$$

$$\begin{aligned} P(X_0 = i, X_1 = j, X_2 = k) &= P(X_2 = k | X_0 = i, X_1 = j) P(X_0 = i, X_1 = j) \\ &= P(X_2 = k | X_1 = j)P(X_0 = i)p_{ij} \\ &= P(X_0 = i)p_{ij}p_{jk} \end{aligned}$$

An equivalent way to state the **Markov condition** is for **any** sequence of states $i, j, k, l, \dots$

$$P(X_0 = i, X_1 = j, X_2 = k, X_3 = l, \dots) = P(X_0 = i)p_{ij}p_{jk}p_{kl} \dots$$

More generally,

Let $\{X_n\}_{n \geq 0}$ be a Markov chain with transition matrix P and initial distribution X_0 is $\vec{\alpha}$.

For any time indices

$$0 \leq n_1 < n_2 < \dots < n_k$$

and any states

$$i_1, i_2, \dots, i_k,$$

the joint distribution of the chain at these times is given by

$$\mathbb{P}(X_{n_1} = i_1, X_{n_2} = i_2, \dots, X_{n_k} = i_k) = (\vec{\alpha} P^{n_1})_{i_1} (P^{n_2 - n_1})_{i_1 i_2} \dots (P^{n_k - n_{k-1}})_{i_{k-1} i_k}$$

Example (Balls and Boxes Markov Chain) .

$$\mathbb{P}(X_0 = 1, X_1 = 2, X_2 = 1, X_3 = 2)$$

Example

Danny's lunch choices from day to day are modeled as a **Markov chain** with four states: Burrito (b), Falafel (f), Pizza (p), and Sushi (s).

The transition matrix P gives the probabilities of what Danny eats tomorrow given what he eats today.

$$P = \begin{pmatrix} 0 & 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0.4 & 0 & 0 & 0.6 \\ 0 & 0.2 & 0.6 & 0.2 \end{pmatrix}$$

On **Sunday** (X_0), Danny has no preference and chooses each lunch option with equal probability $1/4$. That is, the initial distribution for X_0 is

$$\alpha = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right)$$

What is the probability that Danny eats sushi on Wednesday, sushi again on Friday, and pizza on Saturday?

That is

$$\mathbb{P}(X_3 = s, X_5 = s, X_6 = p),$$

Use the Markov property by “cutting” at the specified times:

From Sunday to Wednesday is 3 steps: $\mathbb{P}(X_3 = s) = (\alpha P^3)_s$

From Wednesday (s) to Friday (s) is 2 steps: $\mathbb{P}(X_5 = s \mid X_3 = s) = (P^2)_{ss}$

From Friday (s) to Saturday (p) is 1 step: $\mathbb{P}(X_6 = p \mid X_5 = s) = P_{sp}$.

$$\mathbb{P}(X_3 = s, X_5 = s, X_6 = p) = (\alpha P^3)_s (P^2)_{ss} P_{sp}.$$

$$= 1/4(0.210+0.210+0.360+0.212)*0.4*0.6 \approx 0.0595$$

$$P = \begin{pmatrix} 0 & 0.5 & 0.5 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0.4 & 0 & 0 & 0.6 \\ 0 & 0.2 & 0.6 & 0.2 \end{pmatrix} P^2 = \begin{pmatrix} 0.45 & 0.00 & 0.25 & 0.30 \\ 0.20 & 0.25 & 0.25 & 0.30 \\ 0.00 & 0.32 & 0.56 & 0.12 \\ 0.34 & 0.04 & 0.22 & 0.40 \end{pmatrix} P^3 = \begin{pmatrix} 0.100 & 0.285 & 0.405 & 0.210 \\ 0.225 & 0.160 & 0.405 & 0.210 \\ 0.384 & 0.024 & 0.232 & 0.360 \\ 0.108 & 0.250 & 0.430 & 0.212 \end{pmatrix}$$

❖References:

- MATH7241-Probability 1- Chris King
- Introduction to Probability. C.M. Grinstead and J.L. Snell. American Mathematical Society, 2012
- Introduction to Probability Models 12th Edition by Sheldon M. Ross
- Stochastic Processes 2nd Edition by Sheldon M. Ross
- Probability and Random Processes", G. R. Grimmett and D.R Stirzaker, second edition

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